Nima Yadollahi

Last updated in Oct 2026

Department of physics

Tehran, Iran | yadollahinima83@gmail.com | 0912 545 0533 | Nima Yadollahi | Nima Yadollahi

Education

M.Sc. **Amirkabir University of Technology** - Optics & Photonics — Oct 2026 - Present

B.Sc. **Amirkabir University of Technology** - Physics — Sep 2022 - May 2026

- Last 50 Credit GPA: 18.15/20 | Cumulative GPA: 16.83/20

Honors & Awards

Iranian National M.Sc. Entrance Exam (Konkur Arshad) — Jul 2026

- Ranked 97th of $1,683$ (top 6%) in Physics and 18th of 249 (top 7%) in Photonics

Direct Admission to M.Sc. Program - Amirkabir University of Technology — Jun 2026

- Admitted directly to the Solid State Physics M.Sc. program via the university's talent admission pathway

Experience

Assistant Team Manager – Quantum computing Hackathon (PolyhaQ) — Nov 2025 - Sep 2026

Hackathon Linkedin page: in

- Delivered a coding session about implementing Grover search algorithm | Grover's search implementation
- Collaborated with staff members to organize and host PolyhaQ, Iran's first quantum computing hackathon, to equip passionate students with hands-on quantum computing skills.
- Coordinated with faculty to review competition questions before release and served on the scientific team evaluating and scoring participants' code submissions.
- Certification of contribution as a member of the PolyhaQ Educational Team

Remote Backbend Intern – Achareh Inc. — Jul 2026 - Sep 2026

- I developed an AI agent using the Django framework and an LLM API to assign tasks from transcribed meeting minutes to staff.

Teaching Assistant – Optics — Feb 2026 - May 2026

(Supervisor: Prof. **Parviz Parvin**

- Delivered lectures on basic optics topics including interferometry, diffraction, and polarimetry
- Scored and graded student assignments and exams
- Course Assignments

Teaching Assistant – Quantum Machine Learning — Sep 2025 - Nov 2025

(Supervisor: Asst. Prof. **Negar Ashari Astani**

- Taught fundamental concepts of classical machine learning, including data preprocessing, regression, and random forests.
- Delivered lectures and practical sessions on deep neural network architectures such as MLP, RNN, CNN, and LSTM.
- Course Materials

Certifications

Proficiency in basics Pennylane - Xanadu — Online 2026

- Introduction to Pennylane

School of Quantum Science - Sharif University of Technology — Tehran, Iran 2025

- Quantum Computing Course with Qiskit Library

Advanced Molecular Dynamic and Protein Simulation with GROMACS Tehran, Iran
2025

- GROMACS Course

Publications

Laser-induced fluorescence spectroscopy coupled with physics-aware hybrid machine learning for rapid detection and quantification of hazelnut oil adulteration - Submitted Sep 2026

Nima Yadollahi, Alireza Khalaji, Ali Rahmatpanahi, Ali Bavali

- Developed a machine learning framework for detecting and quantifying hazelnut oil adulteration ($0 - 30\%$) using laser-induced fluorescence (LIF) spectra.
- Designed a compact 1D residual CNN (32K parameters) using eight spectral representations, followed by RBF-kernel SVM/SVR models for classification and concentration prediction.
- Achieved 100.00% excitation-laser , 98.08% adulterant-type, and 91.03% discrete-concentration accuracy; continuous quantification reached MAE = 0.948 pp and $R^2 = 0.958$ on the held-out test set.

Research Experiences

Quantum Boltzmann Machine (Supervisor: Asst. Prof. **Negar Ashari Astani**) Nov 2025 - Present

- Working on generative quantum machine learning models, focusing on the Quantum Boltzmann Machine (QBM), and implemented a QBM training pipeline to streamline its sampling and optimization behavior.

Targeting P-glycoprotein with Cold Atmospheric Plasma-Derived Reactive Species to Overcome Multidrug Resistance (Supervisor: Assoc. Prof. **Hossein Abbasi**) Mar 2024 - Jul 2025

- Analyzed the effects of reactive species (hydrogen peroxide, nitric oxide, and molecular oxygen) on conformational dynamics, revealing significant reductions in flexibility.
- Tools Used: GROMACS, VMD, PyMol

Selected Courses

- Quantum Machine Learning (19.7/20)
- Quantum Computing (20/20)
- Numerical Simulation (17.5/20)
- Electromagnetism I (18.75/20)
- Biophysics (20/20)
- Optics (19.25/20)

Technical Skills

Programming & Machine learning: Python (NumPy, Matplotlib , Pandas), PyTorch, Scikit-learn

Quantum Programming & Simulation: Qiskit, PennyLane, Quantum ESPRESSO, QuTiP, Qiskit Metal, Strawberry Fields

Productivity & Tools: Linux , Git , LaTeX, WordPress, HTML, Microsoft Office

Languages

English: C1 | **French:** A2

References

[1] Department of Physics, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran.

[2] Faculty of Physics, Shahrood University of Technology, Shahrood, Iran.

Negar Ashari Astani, Ph.D.[1]
ashari@aut.ac.ir

Hoshyar Noshad, Ph.D.[1]
hnoshad@aut.ac.ir

Mehdi Khakian, Ph.D.[1]
khakian@aut.ac.ir

Ali Bavali, Ph.D.[1]
alibavali@aut.ac.ir

Saied Setayeshi, Ph.D.[1]
setayesh@aut.ac.ir

Mostafa Annabestani, Ph.D.[2]
annabestani@shahroodut.ac.ir